

TECHSHIP INTERNSHIP

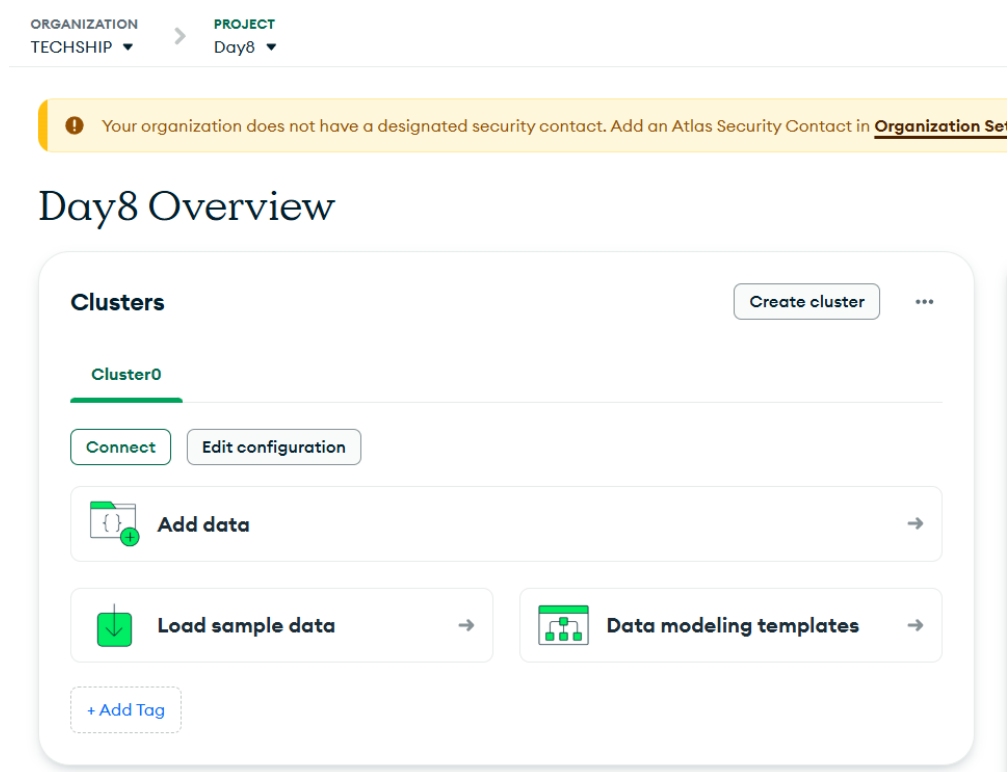
Day 8- MongoDB Fundamentals & Database Operations

Submitted By : Kelna Sebastian

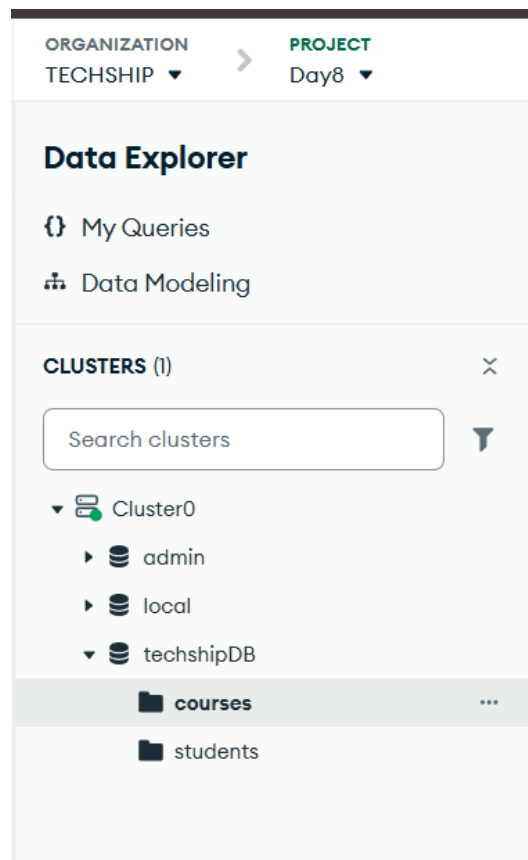
Date : 10-06-2026

During 8th, I gained a practical understanding of databases and their importance in modern applications. I learned how MongoDB stores data using flexible, JSON-like documents and understood the key differences between traditional SQL databases and MongoDB. Working with MongoDB Atlas provided hands-on experience in creating databases and collections, inserting documents, and performing CRUD operations. Initially, navigating the Atlas interface and understanding document structures was slightly challenging. However, through continuous practice, I became more confident in managing and manipulating data effectively. This activity enhanced my understanding of how databases support real-world applications by enabling efficient storage, retrieval, and management of information. Overall, the experience strengthened my foundational knowledge of database concepts and improved my practical skills in using MongoDB.

Activity 1-Create a MongoDB Atlas Account



Activity 2-Create a Database



Activity 3-Insert Student Records

ORGANIZATION: TECHSHIP PROJECT: Day8

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- local
- techshipDB
 - courses
 - students

Cluster0 > techshipDB > students

Documents 5 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA UPDATE DELETE EXPORT CODE

25 1 - 5 of 5

	_id ObjectId	name String	email String	course String	age Int32	
1	ObjectId('6a28fec6ba3d9e0...')	"Kelna Sebastian"	"kelna@example.com"	"Computer Science"	21	Edit Copy Delete
2	ObjectId('6a28fffb3d9e0...')	"Jeeva"	"jeeva@example.com"	"Data Science"	22	Edit Copy Delete
3	ObjectId('6a28ff1fba3d9e0...')	"Anjali"	"anjali@example.com"	"Cyber Security"	20	Edit Copy Delete
4	ObjectId('6a28ff39ba3d9e0...')	"Rahul"	"rahul@example.com"	"AI & ML"	23	Edit Copy Delete
5	ObjectId('6a28ff79ba3d9e0...')	"Jethin Ramdas"	"jethin@example.com"	"Web Development"	25	Edit Copy Delete

Activity 4- Read Data

Cluster0 > techshipDB > students View monitoring Visualize Your Data

Documents 5 Aggregations Schema Indexes 1 Validation

🔍

{ "age": { "\$gt": 23 } }

Generate query ⚡ Explain Reset Find Options ▶

➕ ADD DATA ▾

✎ UPDATE

🗑 DELETE

📄 EXPORT CODE

💡 INSIGHT

25 ▾

1 - 1 of 1

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🏠 students

	_id ObjectId	name String	email String	course String	age Int32	
1	ObjectId('6a28ff79ba3d9e0...	"Jethin Ramdas"	"jethin@example.com"	"Web Development"	25	✎ 🗑 📄

Activity 5-Update Records

Documents 5 Aggregations Schema Indexes 1 Validation

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{ name: "Kelna Sebastian" }

Generate query ⚡ Explain Reset Find Options ▶

➕ ADD DATA ▾

✎ UPDATE

🗑 DELETE

📄 EXPORT CODE

💡

25 ▾

1 - 1 of 1

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🏠 students

	_id ObjectId	name String	email String	course String	age Int32	
1	ObjectId('6a28fec6ba3d9e0...	"Kelna Sebastian"	"kelna@example.com"	"Accounts"	18	✎ 🗑 📄

Activity 6-Delete Records

Cluster0 > techshipDB > students View monitoring Visualize Your Data

Documents 4 Aggregations Schema Indexes 1 Validation

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{ name: "Rahul" }

Generate query ⚡ Explain Reset Find Options ▶

➕ ADD DATA ▾

✎ UPDATE

🗑 DELETE

📄 EXPORT CODE

💡

25 ▾

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Activity 7-Design a Collection

ORGANIZATION
TECHSHIP

PROJECT
Day8

My Queries

Data Modeling

CLUSTERS (1)

Cluster0

admin

local

techshipDB

courses

students

tasks

Welcome

techshipDB

students

tasks

+

Cluster0 > techshipDB > tasks

Documents 5

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

UPDATE

DELETE

EXPORT CODE

25

1 - 5 of 5

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tasks

_id	ObjectId	title	String	description	String	status	String
1	ObjectId('6a2982f0ba3d9e0...')	Submit Work		Upload screenshots and r...		Pending	
2	ObjectId('6a29830aba3d9e0...')	Update Records		Modify student data		Pending	
3	ObjectId('6a298325ba3d9e0...')	Review Notes		Study MongoDB concepts		Completed	
4	ObjectId('6a29833bba3d9e0...')	Prepare Report		Write Internship report		Pending	
5	ObjectId('6a298361ba3d9e0...')	Complete Assignment		Finish MongoDB Activitie...		Pending	

Challenge Activities

Challenge 1-Course Management Collection

My Queries

Data Modeling

CLUSTERS (1)

Cluster0

admin

local

techshipDB

courses

students

tasks

Welcome

techshipDB

students

tasks

courses

+

Cluster0 > techshipDB > courses

Documents 4

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

UPDATE

DELETE

EXPORT CODE

25

1 - 4 of 4

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courses

_id	ObjectId	courseName	String	duration	String	instructor	String	fees	Int32
1	ObjectId('6a2983c3ba3d9e0...')	Mobile App Development		6 Months		Vishnu Raj		18000	
2	ObjectId('6a298483ba3d9e0...')	Java Programming		5 Months		Priyadarshani Ramdas		14000	
3	ObjectId('6a298438ba3d9e0...')	Python Programming		4 Months		Anjali Nair		12000	
4	ObjectId('6a298444ba3d9e0...')	Web Development		6 Months		Arun Kumar		15000	

Challenge 2-Employee Database

Cluster0 > techshipDB > employees

Documents 5

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

UPDATE

DELETE

EXPORT CODE

25

1 - 5 of 5

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employees

_id	ObjectId	employeeName	String	department	String	salary	Int32	experience	Int32
1	ObjectId('6a2984afba3d9e0...')	Vishnu		IT		45000		5	
2	ObjectId('6a2984d2ba3d9e0...')	Anu		Marketing		32000		2	
3	ObjectId('6a2984ebba3d9e0...')	Rahul		Finance		40000		4	
4	ObjectId('6a298508ba3d9e0...')	Meera		HR		30000		3	
5	ObjectId('6a298513ba3d9e0...')	Akhil		IT		35000		2	

Create

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

local

techshipDB

courses

employees

Welcome

techshipDB

employees

students

tasks

courses

+

Cluster0 > techshipDB > employees

View monitoring

Visualize Your Data

Documents 5

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

Generate query

Explain

Reset

Find

Options

ADD DATA

UPDATE

DELETE

EXPORT CODE

25

1 - 2 of 2

employees

	_id ObjectId	employeeName String	department String	salary Int32	experience Int32	
1	ObjectId('6a2984d2ba3d9e0...')	Anu	Marketing	32000	2	
2	ObjectId('6a298513ba3d9e0...')	Akhil	IT	35000	2	

Read

Documents 5

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

UPDATE

DELETE

EXPORT CODE

25

1 - 5 of 5

employees

	_id ObjectId	employeeName String	department String	salary Int32	experience Int32	
1	ObjectId('6a2984afba3d9e0...')	Vishnu	IT	45000	5	
2	ObjectId('6a2984d2ba3d9e0...')	Anu	Marketing	32000	2	
3	ObjectId('6a2984ebba3d9e0...')	Rahul	Finance	40000	4	
4	ObjectId('6a298500ba3d9e0...')	Meera	HR	30000	3	
5	ObjectId('6a298513ba3d9e0...')	Akhil	IT	35000	10	

Update

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

local

techshipDB

courses

employees

students

tasks

Welcome

techshipDB

employees

students

tasks

courses

+

Cluster0 > techshipDB > employees

View monitoring

Visualize Your Data

Documents 4

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

UPDATE

DELETE

EXPORT CODE

25

1 - 4 of 4

employees

	_id ObjectId	employeeName String	department String	salary Int32	experience Int32	
1	ObjectId('6a2984afba3d9e0...')	Vishnu	IT	45000	5	
2	ObjectId('6a2984d2ba3d9e0...')	Anu	Marketing	32000	2	
3	ObjectId('6a298508ba3d9e0...')	Meera	HR	30000	3	
4	ObjectId('6a298513ba3d9e0...')	Akhil	IT	35000	10	

Delete

Challenge 3-Design Your Own Database

Library Management System

The screenshot shows the Data Explorer interface with the 'books' collection selected. The left sidebar shows the cluster hierarchy: Cluster0 > libraryDB > books. The main panel displays the 'books' collection with a table of documents. The table has columns: _id, ObjectID, title, author, genre, and available. There are 3 documents listed.

	_id	ObjectID	title	author	genre	available
1		ObjectID('6a29074bba3d9e0...')	"The Alchemist"	"Paulo Coelho"	"Fiction"	true
2		ObjectID('6a29076eba3d9e0...')	"Atomic Habits"	"James Clear"	"Self Help"	true
3		ObjectID('6a2907b4ba3d9e0...')	"Harry Potter and the Phi..."	"J.K. Rowling"	"Fantasy"	false

The screenshot shows the Data Explorer interface with the 'members' collection selected. The left sidebar shows the cluster hierarchy: Cluster0 > libraryDB > members. The main panel displays the 'members' collection with a table of documents. The table has columns: _id, ObjectID, name, email, and membershipType. There are 3 documents listed.

	_id	ObjectID	name	email	membershipType
1		ObjectID('6a2907f5ba3d9e0...')	"Kelna Sebastian"	"kelna@example.com"	"Student"
2		ObjectID('6a290829ba3d9e0...')	"Joseph Mathew"	"joseph@example.com"	"Faculty"
3		ObjectID('6a290845ba3d9e0...')	"Jeeva"	"jeeva@example.com"	"Student"

Assignment

Student Management Database

Database Name: studentManagement

Collections Created:

- students
- courses

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- libraryDB
 - books
 - members
- local
 - oplog.rs
- studentManagement
 - courses**
 - students

Cluster0 > studentManagement > courses

Documents 5 Aggregations Schema Indexes 1 Validation

Generate query + Explain Reset Find Options

ADD DATA UPDATE DELETE EXPORT CODE

25 1 - 5 of 5

	_id ObjectId	courseId String	courseName String	duration String	instructor String	
1	ObjectId('6a298a92ba3d9e0...')	"C001"	"Computer Science"	"4 Years"	"Dr. Sharma"	
2	ObjectId('6a298ac1ba3d9e0...')	"C002"	"Web Development"	"6 Months"	"Ms. Nair"	
3	ObjectId('6a298aebba3d9e0...')	"C003"	"Data Science"	"1 Year"	"Mr. Thomas"	
4	ObjectId('6a298b1fba3d9e0...')	"C004"	"Cyber Security"	"8 Months"	"Ms. Joseph"	
5	ObjectId('6a298b32ba3d9e0...')	"C005"	"AI & ML"	"1 Year"	"Dr. Menon"	

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- libraryDB
- local
- studentManagement
 - courses
 - students**
- techshipDB

Cluster0 > studentManagement > students

Documents 10 Aggregations Schema Indexes 1 Validation

Generate query + Explain Reset Find Options

ADD DATA UPDATE DELETE EXPORT CODE

25 1 - 10 of 10

	_id ObjectId	studentId String	name String	email String	course String	
1	ObjectId('6a29894eba3d9e0...')	"S001"	"KeIra Sebastian"	"keIra@example.com"	"ACCA"	
2	ObjectId('6a298967ba3d9e0...')	"S002"	"Alvin Joseph"	"alvin@example.com"	"Web Development"	
3	ObjectId('6a29898cba3d9e0...')	"S003"	"Jeeva"	"jeeva@example.com"	"Data Science"	
4	ObjectId('6a2989b3ba3d9e0...')	"S004"	"Anjali"	"anjali@example.com"	"Cyber Security"	
5	ObjectId('6a2989d3ba3d9e0...')	"S005"	"Rahul"	"rahul@example.com"	"AI & ML"	
6	ObjectId('6a2989f3ba3d9e0...')	"S006"	"Neha"	"neha@example.com"	"Cloud Computing"	
7	ObjectId('6a298a13ba3d9e0...')	"S007"	"Arjun"	"arjun@example.com"	"Software Engineering"	
8	ObjectId('6a298a2eba3d9e0...')	"S008"	"Priya"	"priya@example.com"	"UI/UX Design"	
9	ObjectId('6a298a4aba3d9e0...')	"S009"	"Vishnu"	"vishnu@example.com"	"Machine Learning"	

Create: Created the studentManagement database and the students and courses collections.
Inserted 10 student records and 5 course records.

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- libraryDB
 - books
 - members
- local
 - oplog.rs
- studentManagement
 - courses**
 - students

Cluster0 > studentManagement > courses

Documents 5 Aggregations Schema Indexes 1 Validation

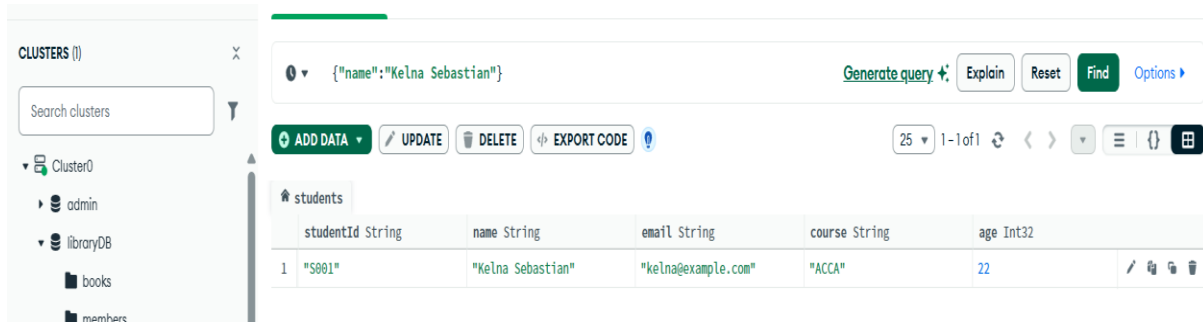
Generate query + Explain Reset Find Options

ADD DATA UPDATE DELETE EXPORT CODE

25 1 - 5 of 5

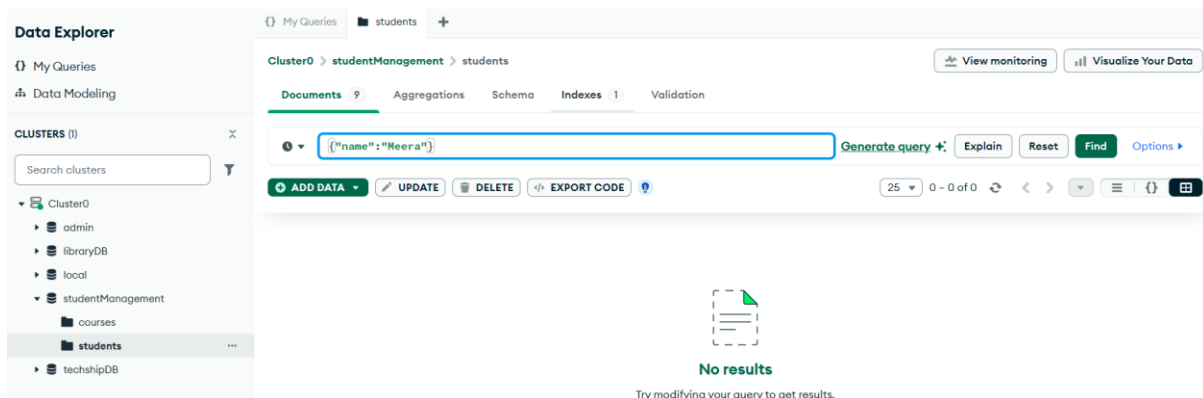
	_id ObjectId	courseId String	courseName String	duration String	instructor String	
1	ObjectId('6a298a92ba3d9e0...')	"C001"	"Computer Science"	"4 Years"	"Dr. Sharma"	
2	ObjectId('6a298ac1ba3d9e0...')	"C002"	"Web Development"	"6 Months"	"Ms. Nair"	
3	ObjectId('6a298aebba3d9e0...')	"C003"	"Data Science"	"1 Year"	"Mr. Thomas"	
4	ObjectId('6a298b1fba3d9e0...')	"C004"	"Cyber Security"	"8 Months"	"Ms. Joseph"	
5	ObjectId('6a298b32ba3d9e0...')	"C005"	"AI & ML"	"1 Year"	"Dr. Menon"	

Read: Retrieved and viewed all student documents using the Find operation



Update

Update: Updated a student's information by modifying the age field and verified the changes.



Delete

Delete: Deleted one student document from the collection and confirmed its successful removal.

Conclusion

The studentManagement database was successfully created with two collections: students and courses. Sample documents were added to demonstrate the structure of each collection, and key CRUD operations were performed to manage the data effectively. The Read operation allowed retrieval of student records, while the Delete operation confirmed the ability to remove documents when required. This exercise provided practical experience in working with MongoDB collections and reinforced the fundamental concepts of database management and document manipulation.